The Power of Vaccination

From Smallpox Eradication to COVID-19 in Record Time

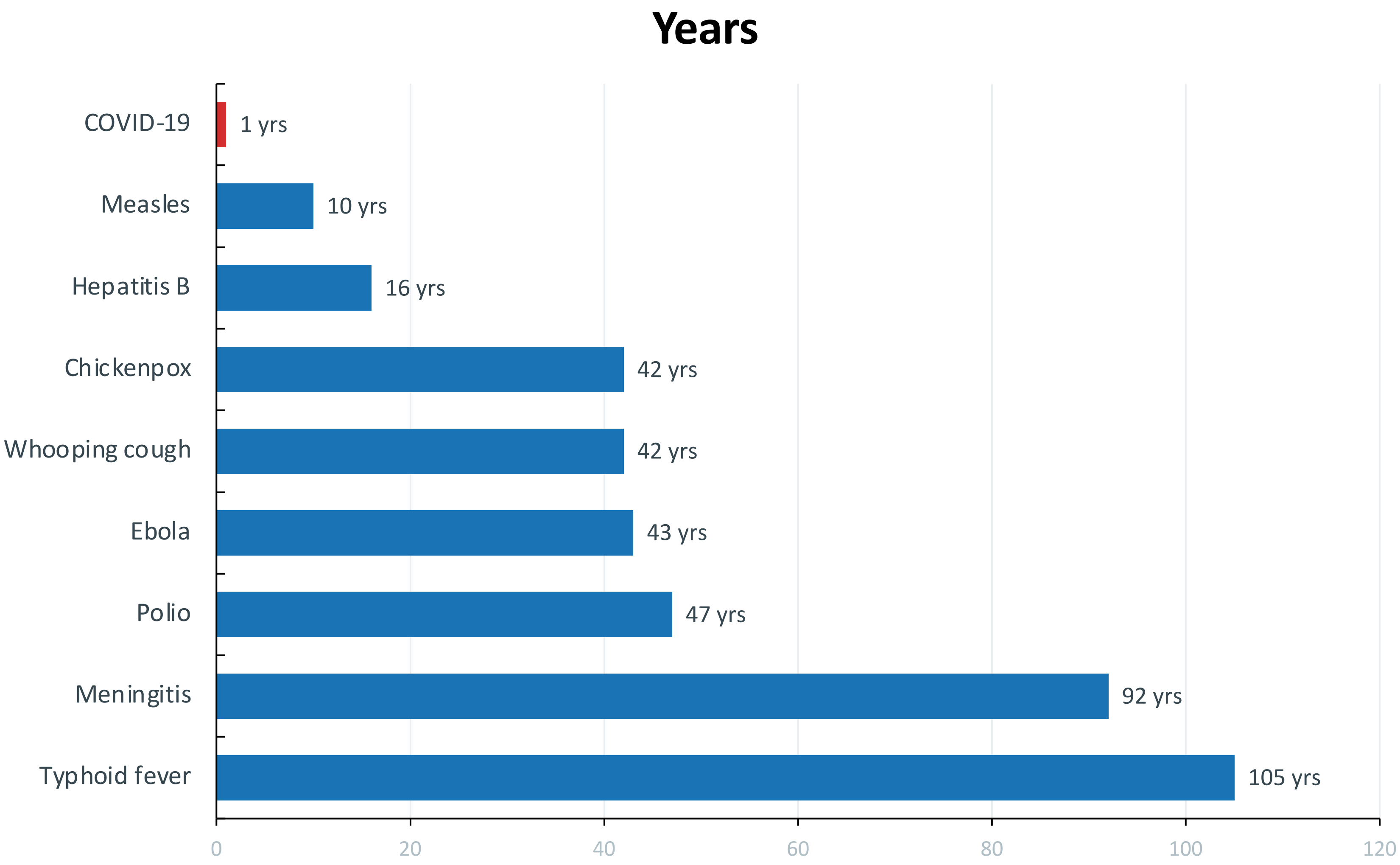
A data-driven overview of how vaccines have transformed public health
by eliminating or drastically reducing deadly diseases over the past two centuries.

Spanning 1796 – 2020 | For Public Health Professionals & Policy Analysts

Source: Our World in Data — ourworldindata.org/vaccination

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

Time from pathogen identification to vaccine licensure



COVID-19

< 1 Year

Enabled by mRNA technology
and genome sequencing

KEY INSIGHT

Typhoid took 105 years.
COVID-19 took under 1 year.
A 100× acceleration.

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox

Pre-vaccine vs. post-vaccine annual cases and deaths per million in the US — Source: Roush & Murphy (2007)

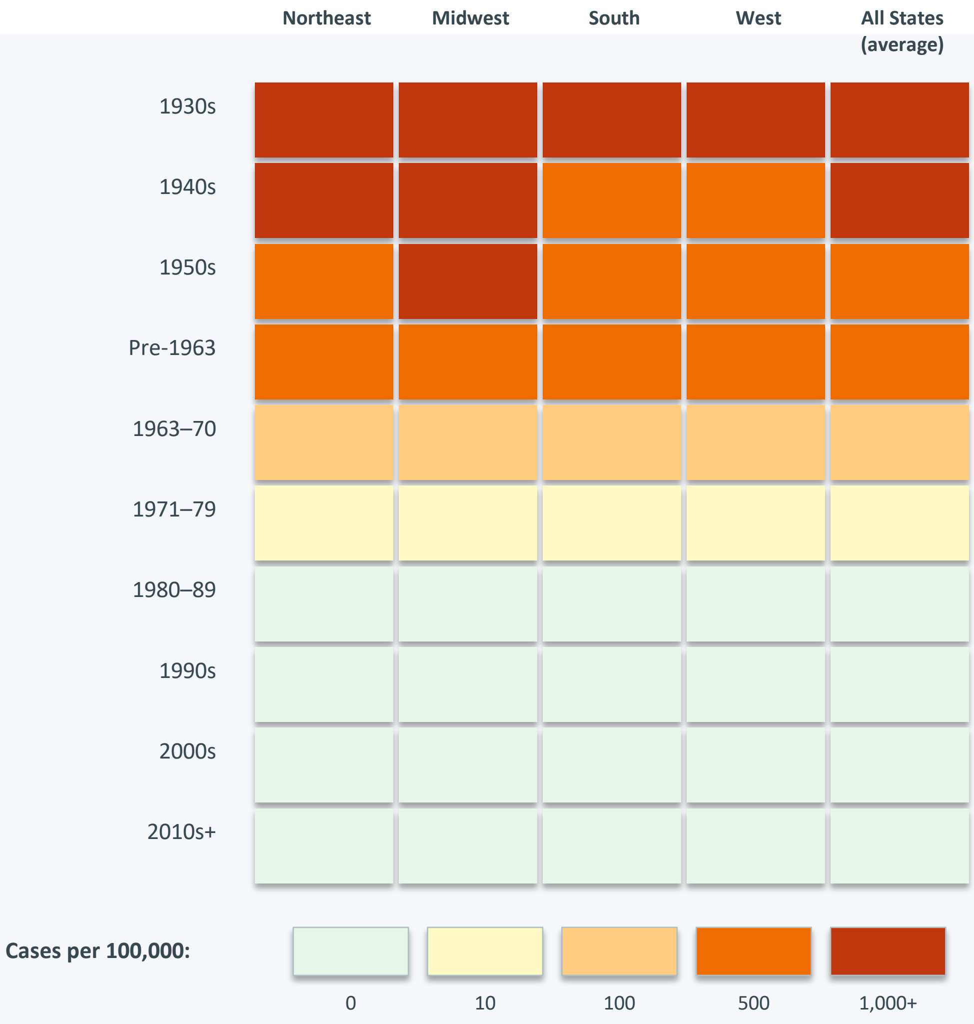
Disease	Pre-Vaccine Cases/M/yr	Post-Vaccine Cases/M/yr	Case Reduction	Pre-Vaccine Deaths/M/yr	Death Reduction
Diphtheria	158	0	**100%**	13.7	**100%**
Smallpox	250	0	**100%**	2.9	**100%**
Acute Polio	141	0	**100%**	10	**100%**
Measles	3,044	0.2	99.99%	2.5	100%
Rubella	242	0.04	99.98%	0.09	100%
Pertussis	1,534	52	96.6%	30.8	99.7%
Hepatitis A	465	51	**89%**	0.5	**88.7%**
Hepatitis B	273	44	**83.9%**	1	**83.6%**
Pneumococcal	233	139	**40.5%**	24	**31.3%**

= 100% reduction achieved

= Incomplete control — gaps persist

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

Measles cases per 100,000 by state, 1929–2022 — heatmap visualization



1963 First measles vaccine developed by John Enders

1971 MMR vaccine developed by Maurice Hilleman

1980 Vaccination mandatory for kindergarten entry

After the 1980 mandate, cases dropped to near zero across all states, with only sporadic outbreaks.

Smallpox: The Only Human Disease Eradicated Through Vaccination

A global campaign spanning two decades — from WHO resolution to worldwide certification

- **1796** — Jenner creates first vaccine (cowpox)
- **1959** — WHO passes eradication resolution
- **1967** — Intensified global campaign
- **1977** — Last natural case (Somalia)
- **1980** — Eradication certified worldwide

WHY SMALLPOX WAS ERADICABLE

- Virus infected only humans — no animal reservoir
- Symptoms were clearly distinguishable
- Vaccine was cheap and highly effective

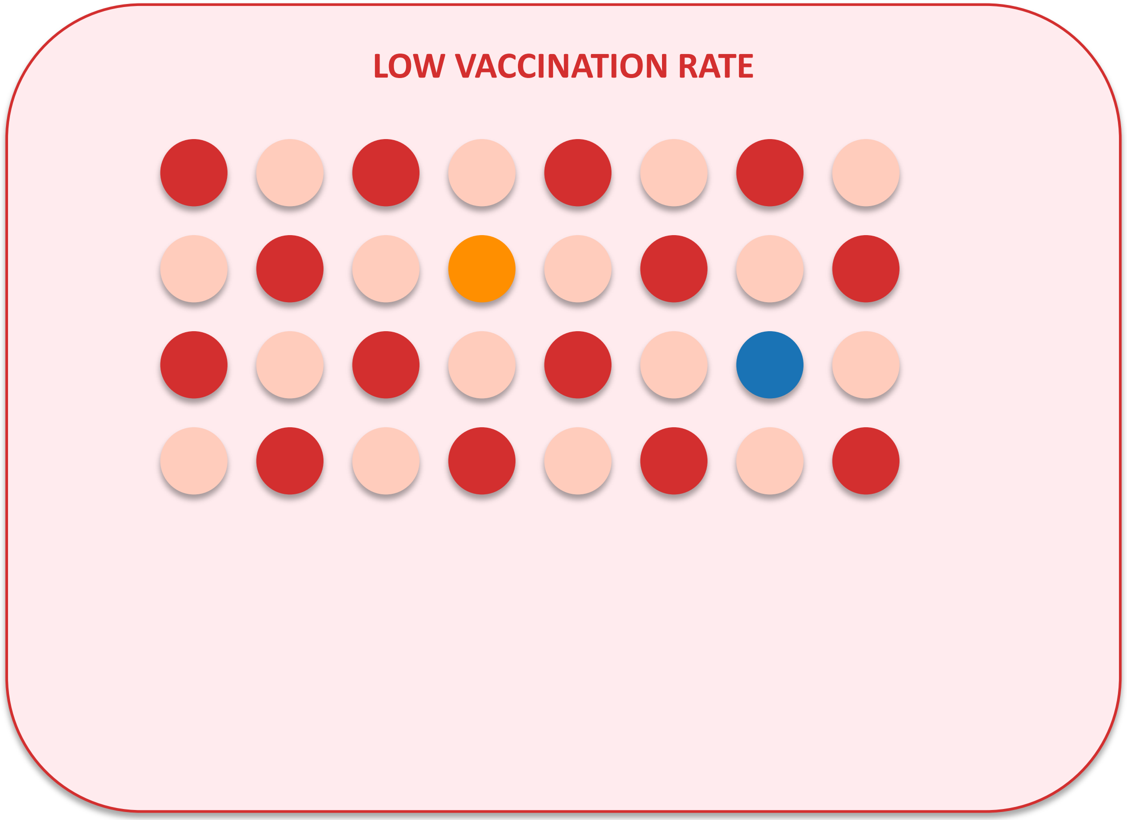
ACHIEVEMENT

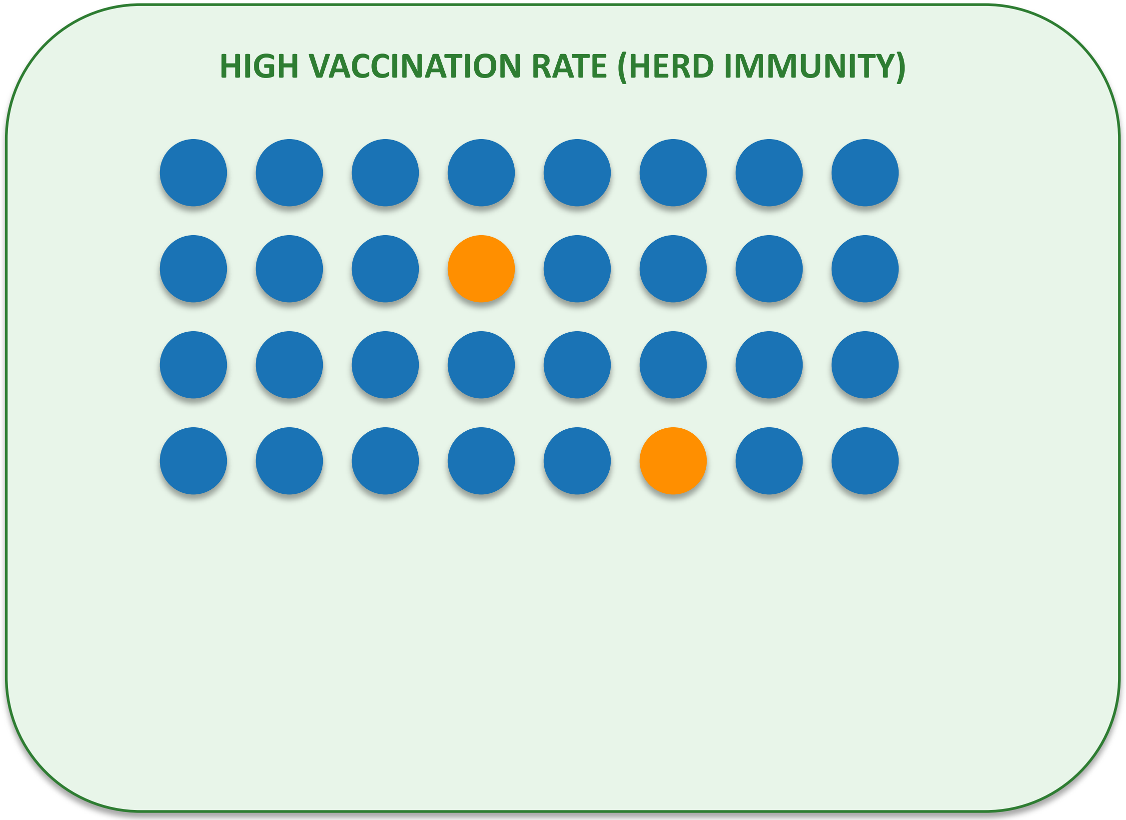
Smallpox remains the only human disease fully eliminated through vaccination.

Pre-vaccine: 250 cases per million/yr in the US
Post-vaccine: 0 cases — 100% elimination

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

When enough people are vaccinated, pathogens cannot spread effectively





- Vaccinated
- Infected
- Unvaccinated
- Vulnerable (cannot be vaccinated)

Vulnerable populations include newborns, immunocompromised individuals, and cancer patients on chemotherapy who cannot be vaccinated.

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

Incomplete vaccine impact and barriers to global coverage

Pneumococcal Disease

40.5%
case reduction

31.3%
death reduction

Hepatitis A

89%
case reduction

88.7%
death reduction

Acute Hepatitis B

83.9%
case reduction

83.6%
death reduction

BARRIERS TO PROGRESS

Access Gaps

Not everyone globally has access to vaccines

Supply Shortages

Insufficient supply puts children at risk

Trust Challenges

Poor scientific understanding reduces uptake

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

1. **100% Elimination Achieved**
   Diphtheria, smallpox, and polio cases and deaths fully eliminated in the US through vaccination.

2. **Development Timelines Collapsed**
   From 105 years (typhoid) to under 1 year (COVID-19), driven by mRNA technology and genomic tools.

3. **Measles Near Zero**
   Cases dropped to near zero across all 50 US states after the 1980 kindergarten vaccination mandate.

4. **Herd Immunity Works**
   Population-level vaccination shields newborns, immunocompromised individuals, and chemotherapy patients.

5. **Gaps Persist**
   Pneumococcal disease (40.5% case reduction), hepatitis A (89%), and hepatitis B (83.9%) need improved coverage.

CALL TO ACTION

- Expand access to vaccines in underserved regions globally
- Strengthen supply chains to prevent shortages
- Invest in next-generation vaccines for diseases with incomplete control
- Build public trust through transparent science communication
- Sustain funding for global eradication and elimination programs

Source: Our World in Data — ourworldindata.org/vaccination | Data: Roush & Murphy (2007)